WORKED SOLUTION

# STEP III 1995, Question 7

Pure • Groups and Isomorphisms

**STEP III 1995, Question 7**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q7

## The key idea

Associativity is given, so for each set there are only three things to check: closure, an identity, and an inverse for every element. A set fails as soon as one of them breaks, and a single explicit counterexample is enough. For the isomorphisms, look for structural features that any isomorphism must preserve: the number of elements, and the orders of the elements.

## Solution

Throughout, "multiplication" means ordinary multiplication of complex numbers, multiplication modulo $n$, or matrix multiplication, as appropriate.

### Part (i)

Let $S = \{z \in \mathbb{C} : |z| = 1\}$.

- *Closure:* if $|z| = |w| = 1$ then $|zw| = |z||w| = 1$.
- *Identity:* $1 \in S$ and $1z = z1 = z$.
- *Inverses:* if $|z| = 1$ then $z \neq 0$, and $z^{-1} = \bar{z}$ (since $z\bar{z} = |z|^2 = 1$), which also has modulus 1.

So $S$ is a group (an infinite one).

### Part (ii)

Under multiplication modulo 4 the set $\{0, 1, 2, 3\}$ is closed and has identity 1, but 0 has no inverse, since $0 \times b \equiv 0 \not\equiv 1$ for every $b$. So it is *not* a group. (Removing 0 doesn't rescue it: 2 still has no inverse, as $2b$ is always even, and in fact $2 \times 2 \equiv 0$, so $\{1, 2, 3\}$ isn't even closed.)

**Part (iii)**

Using the compound-angle formulae,

$$\mathrm{M}(\alpha)\mathrm{M}(\beta) = \begin{pmatrix} c_\alpha c_\beta - s_\alpha s_\beta & -(s_\alpha c_\beta + c_\alpha s_\beta) \\ s_\alpha c_\beta + c_\alpha s_\beta & c_\alpha c_\beta - s_\alpha s_\beta \end{pmatrix} = \mathrm{M}(\alpha+\beta),$$

writing $c_\alpha = \cos\alpha$, $s_\alpha = \sin\alpha$ and so on. Since cos and sin have period $2\pi$, $\mathrm{M}(\alpha+\beta) = \mathrm{M}(\gamma)$ where $\gamma$ is $\alpha+\beta$ reduced to lie in $[0, 2\pi)$. So:

- *Closure:* the product of two matrices in the set is again in the set.
- *Identity:* $\mathrm{M}(0) = I$.
- *Inverses:* $\mathrm{M}(\theta)\mathrm{M}(2\pi-\theta) = \mathrm{M}(2\pi) = I$ for $0 < \theta < 2\pi$, and $\mathrm{M}(0)$ is its own inverse.

So this set is a group.

**Part (iv)**

Modulo 8,

$$3^2 = 9 \equiv 1, \qquad 5^2 = 25 \equiv 1, \qquad 7^2 = 49 \equiv 1,$$
$$3\times 5 = 15 \equiv 7, \qquad 3\times 7 = 21 \equiv 5, \qquad 5\times 7 = 35 \equiv 3.$$

With the trivial products involving 1, this gives the whole multiplication table (multiplication is commutative), and every product lies in $\{1,3,5,7\}$, so the set is closed. The identity is 1, and each element is its own inverse. So this is a group, of order 4, in which every element other than 1 has order 2.

**Part (v)**

The identity matrix $I$ has integer entries, and the set is closed (products and sums of integers are integers), but inverses fail. The zero matrix has no inverse at all. Even an invertible matrix can fail: $A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ has inverse $\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix}$ in the real matrices, and inverses are unique, so $A$ has no inverse with integer entries. So this set is *not* a group.

> **Ask yourself** Is pointing at the zero matrix really enough, or do we need to say something about all the matrices?

One element without an inverse is enough to show that the set is not a group. The second example shows that the problem goes deeper than singular matrices. An integer matrix has an integer inverse exactly when its determinant is $\pm 1$, because $\det A \det A^{-1} = 1$ with both determinants integers. Restricting to those matrices does give a group.

#### Part (vi)

Modulo 5, the powers of 2 are

$$2^1 = 2, \quad 2^2 = 4, \quad 2^3 = 8 \equiv 3, \quad 2^4 = 16 \equiv 1.$$

So $\{1,2,3,4\} = \{2^0, 2^1, 2^2, 2^3\}$, and $2^a 2^b = 2^{a+b}$ with exponents read modulo 4. Hence the set is closed, 1 is the identity, and $2^a$ has inverse $2^{4-a}$. It is a group: the cyclic group of order 4, generated by 2 (or by 3).

#### Isomorphisms

The groups are (i), (iii), (iv) and (vi), which gives six pairs.

**(i) and (iii) are isomorphic.** Every unit complex number can be written uniquely as $e^{i\theta}$ with $0 \leq \theta < 2\pi$. Define $\phi(e^{i\theta}) = M(\theta)$. This is a bijection, since distinct $\theta$ in $[0, 2\pi)$ give distinct matrices (the first column $(\cos\theta, \sin\theta)$ determines $\theta$). It respects multiplication: $e^{i\alpha}e^{i\beta} = e^{i(\alpha+\beta)}$ and $M(\alpha)M(\beta) = M(\alpha+\beta)$, where in both groups the angle is reduced modulo $2\pi$. Geometrically, both groups are the rotations of the plane about the origin.

**(i) or (iii) with (iv) or (vi): not isomorphic.** An isomorphism is a bijection, so the two groups must have the same number of elements. Groups (i) and (iii) are infinite, while (iv) and (vi) have four elements each. That disposes of four pairs.



**Ask yourself** (iv) and (vi) both have four elements and both are commutative. What could tell them apart?

**(iv) and (vi): not isomorphic.** An isomorphism $\phi$ preserves the order of every element: $\phi(g)^n = \phi(g^n)$, which equals the identity exactly when $g^n$ is the identity, because $\phi$ is injective and sends identity to identity. In (vi) the element 2 has order 4. In (iv) every element $g$ satisfies $g^2 = 1$, so no element has order 4. So no isomorphism exists.

In summary, the only isomorphic pair is (i) and (iii).

## Checks

A computer confirmed the closure, identity and inverse properties for (iv) and (vi) by brute force over the tables. It found that 0 and 2 have no inverse modulo 4, and that the element orders are $1,2,2,2$ in (iv) and $1,4,4,2$ (for $1,2,3,4$) in (vi). It also checked $M(\alpha)M(\beta) = M(\alpha+\beta)$ and the correspondence with $e^{i\theta}$ at $10^4$ random angles. You can check (iv) quickly by hand: squaring any odd number gives 1 modulo 8, since $(2k+1)^2 = 4k(k+1)+1$ and $k(k+1)$ is even.

## Common slips

- Saying (iv) and (vi) are isomorphic "because both have four elements". Equal size is necessary but not sufficient.

- In (v), checking only that $\det A \neq 0$. The inverse must also have integer entries.
- Reading (ii) as addition modulo 4. Under addition it would be a cyclic group of order 4, isomorphic to (vi), but the question specifies multiplication.

## Beyond STEP

Up to isomorphism there are exactly two groups of order 4: the cyclic group $C_4$, as in (vi), and the Klein four-group $C_2 \times C_2$, as in (iv). Classifying small groups by element orders is one of the first things a university group theory course does. The circle group of (i) and (iii) is the simplest example of a Lie group, a group that is also a smooth curve or surface. The integer matrices of determinant $\pm 1$ from (v) form the group $\mathrm{GL}_2(\mathbb{Z})$, which is important in number theory. Allenby's *Rings, Fields and Groups* is a good place to read more.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.